

Assignment 3

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1 Introduction

The objective of this assignment is to simulate the diffusion of information through a social network where different types of actors coexist. The network represents interactions between anime entries extracted from the MyAnimeList dataset. Each node corresponds to an anime, while edges represent similarity relationships based on shared genres. Three different kinds of information are considered:

- a reliable message propagated by Verifiers;
- a mutated message generated by Unintentional Spreaders;
- fake information introduced by Social Bots.

The goal is to analyze how heterogeneous node behaviors influence the diffusion process and how misinformation may propagate through the network.

2 Actor Roles

The simulation includes four types of nodes, each associated with a specific message behavior. Table 1 summarizes their characteristics.

Role	Behavior	Message
Verifier	Reliable source that always broadcasts the correct information and never changes state.	Real
Unintentional Spreader	Accidentally alters the true message once upon adoption and forwards the modified version.	Altered
Normal Node	Adopts a message if enough neighbors support it (threshold rule: $\geq \theta$).	Majority or inactive
Social Bot	Replaces any received message with fake information and spreads it aggressively.	Fake

Table 1: Summary of node roles and message behaviors.

3 Diffusion Model

The spreading process evolves in discrete iterations. At every iteration:

1. Verifiers keep broadcasting the true message.
2. Unintentional nodes may transform the true message into the mutated one.
3. Normal nodes adopt the majority message if the threshold condition is satisfied.
4. Bots replace every received message with fake information.

The simulation terminates when the network reaches a stable state. A maximum of 20 iterations is used as a safety limit.

4 Preliminary Validation on a Small Graph

Before applying the diffusion model to the full anime network, a preliminary experiment was conducted on a small synthetic graph in order to validate the correctness of the implementation and to observe the expected qualitative behavior of the model.

The small graph consisted of 10 nodes with random connectivity. To evaluate the sensitivity of the model, the number of Unintentional Spreaders was progressively increased, and the resulting fraction of mutated messages was recorded.

Figure 1 shows the mutation curve obtained from the small graph.

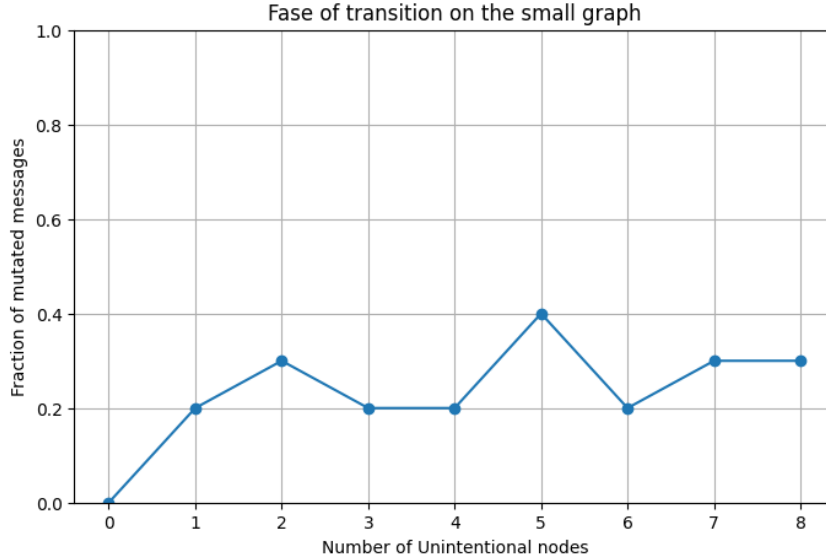


Figure 1: Fraction of mutated messages as the number of Unintentional Spreaders increases in the small graph.

With only 10 nodes, even a single mutated adoption corresponds to 10% of the network, making the curve extremely sensitive to small structural variations. Moreover, the position of Unintentional Spreaders plays a dominant role: when an Unintentional node lies on a central path, the mutated message can reach several neighbors, producing peaks such as 0.4 for five Unintentional nodes. Conversely, when Unintentional nodes are placed in peripheral regions, the mutated message fails to propagate, resulting in lower values.

This volatility is a characteristic of small random graphs, where the threshold condition is easily satisfied and local structures strongly influence the diffusion outcome. These preliminary results confirm that the model behaves correctly and highlight the importance of network size and connectivity in stabilizing the diffusion process.

5 Diffusion on the Anime Network

After validating the model on a small synthetic graph, the diffusion process was applied to the full anime similarity network. To understand which structural and behavioral factors influence the spread of mutated information, four controlled experiments were performed. Each experiment isolates a specific mechanism of the model: random placement, structural importance, threshold sensitivity, and verifier reinforcement.

Figure 2 shows the mutation curve obtained from the anime network.

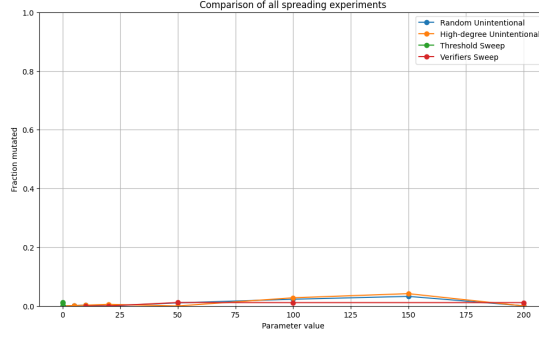


Figure 2: Fraction of mutated messages as the number of Unintentional Spreaders increases in the anime network.

5.1 Random Unintentional Spreaders

The first experiment evaluates how accidental misinformation behaves when Unintentional nodes are placed uniformly at random. The mutated fraction remains extremely low across all tested configurations (from 0% to only 4.36%). This happens because random placement rarely positions Unintentional nodes in structurally relevant regions of the graph. Most of them fall in peripheral or redundant neighborhoods, where their influence is immediately overwhelmed by the true message broadcasted by Verifiers and reinforced by the majority rule applied by Normal nodes.

5.2 High-Degree Unintentional Spreaders

To test whether structurally important nodes can amplify mutation, Unintentional Spreaders were selected among the highest-degree nodes. The mutated fraction increases slightly compared to random placement, reaching a maximum of 7.18% when 200 high-degree nodes are used. High-degree nodes have many neighbors and could in principle spread the mutated message widely. However, their neighborhoods are typically dominated by Verifiers or nodes already aligned with the true message. As a result, even structurally central Unintentional nodes fail to trigger large-scale mutation.

5.3 Threshold Sweep

The threshold parameter controls how easily Normal nodes adopt a message. Sweeping this value reveals a sharp transition: for very low thresholds (0.01–0.05), a small amount of mutation appears (around 1%), but once the threshold reaches 0.1, the mutated message disappears entirely. This shows that even a modest adoption requirement is sufficient to prevent accidental misinformation from spreading in a dense network. The high connectivity of the anime graph ensures that the true message almost always constitutes the local majority, making mutation unable to surpass the threshold.

5.4 Verifiers Sweep

Finally, the number of Verifiers was varied to measure how strongly they stabilize the diffusion process. The mutated fraction remains consistently low regardless of whether Verifiers are few (10) or many (200), always around 1%. This indicates that the network’s robustness does not depend primarily on the number of reliable sources: the structural properties of the graph already ensure that the true message dominates the diffusion process. Additional Verifiers simply reinforce an already stable configuration.

5.5 Summary

Across all four experiments, the mutated message remains confined to a very small portion of the network, demonstrate that the anime similarity graph is highly resilient to accidental misinformation.

6 Diffusion with Social Bots

In the second part of the assignment, the diffusion model is extended by introducing Social Bots, i.e., malicious actors designed to actively subvert the flow of information. Unlike Unintentional Spreaders, which generate misinformation only by mistake, Bots deliberately replace any received message with fake news and propagate it aggressively to all their neighbors. Their behavior is therefore substantially more disruptive than accidental mutation.

6.1 Bot Mechanics

Bots follow two strengthened behavioral rules:

- **Active subversion:** as soon as a Bot receives any message (true, mutated, or fake), it immediately replaces it with fake news and broadcasts it to all neighbors.
- **Zero-threshold activation:** a single active neighbor is sufficient to trigger fake propagation. Bots do not rely on the adoption threshold used by Normal nodes.

To maximize their disruptive potential, Bots are placed strategically. Specifically, they are selected among the nodes with highest betweenness centrality, ensuring that they occupy structurally influential positions in the network.

6.2 Simulation Results

The diffusion process was executed twice: once using the baseline configuration of Part 1 (without Bots), and once after introducing three strategically placed Bots. Figure 3 shows the final state of the network in both scenarios.

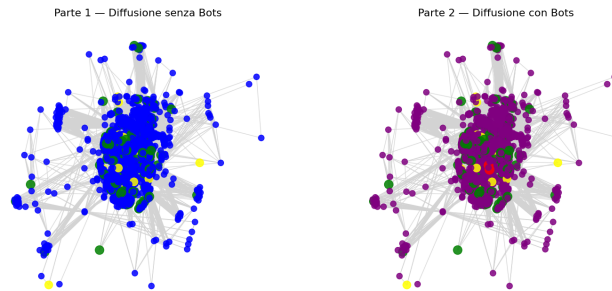


Figure 3: Comparison between diffusion without Bots (left) and with Bots (right).

Node Role	Color	Message State	Color
Verifier	Green	True (Real)	Blue
Unintentional Spreader	Yellow	Mutated (Altered)	Orange
Social Bot	Red	Fake (Fake)	Purple
		None	Gray

Table 2: Color encoding used in Figure 3 for node roles (left) and message states (right).

In the left panel (diffusion without Bots), most nodes appear in blue, indicating that the true message successfully spreads across the network. Accidental mutation is extremely

rare (only 1.2% of nodes), and the few mutated nodes are visually negligible in the layout, which is why almost no orange nodes are visible. As expected, no purple nodes (Fake) are present.

In the right panel (diffusion with Bots), the network becomes dominated by purple nodes. Despite being few and strategically placed, Bots trigger a cascade of fake-news adoption that overwhelms the network. The collapse of blue nodes illustrates how malicious actors override both correct information and accidental mutation.

Overall, the visualization highlights the dramatic contrast between the two scenarios: the network is robust to accidental errors but highly vulnerable to strategically placed malicious Bots. The quantitative results are summarized below:

Scenario	True (%)	Mutated (%)	Fake (%)	None (%)
Without Bots	79.21	1.20	0.00	19.58
With Bots	5.61	0.00	73.60	20.79

Table 3: Final message distribution with and without Social Bots.

6.3 Discussion

In the absence of Bots, the anime similarity network exhibits strong resilience: the true message dominates the diffusion process, accidental mutation remains marginal, and no fake information spreads. This confirms the robustness observed in Part 1.

The introduction of only three Bots, however, completely overturns the dynamics. Fake information becomes the dominant message, reaching more than 73% of the nodes, while the true message collapses to just 5.6%. The mutated message disappears entirely, as Bots overwrite any message they encounter.

This dramatic shift highlights a key insight: the network is robust to accidental errors but highly vulnerable to malicious actors placed in structurally central positions. Bots exploit their strategic location and zero-threshold activation to rapidly contaminate the network, demonstrating how even a small number of adversarial nodes can destabilize an otherwise stable diffusion process.

6.4 Summary

Social Bots represent a qualitatively different threat compared to Unintentional Spreaders. While accidental mutation remains confined and easily suppressed, malicious fake-news propagation quickly becomes dominant when Bots are introduced. This experiment illustrates the importance of node behavior and structural centrality in determining the stability of information flow within complex networks.

7 Final Conclusion

The results reveal a clear and meaningful distinction between accidental misinformation and intentional fake-news propagation.

The network is robust to accidental mutation but highly vulnerable to coordinated malicious behavior. The structural centrality of Bots amplifies their influence, enabling them to override the natural stability of the diffusion process.

Overall, these findings highlight how a small number of strategically positioned malicious actors can significantly compromise the reliability of information diffusion.